

Research on Fault Line Selection in a Resonant Grounded System Based on the Instantaneous Power Theory

* Li-Qun Shang and Xing Wu

Abstract

A new method of composite line selection based on the instantaneous power is proposed in this paper. The steady state and transient state fault characteristics of single-phase grounding faults are analyzed. Firstly, the harmonic is extracted by the instantaneous power theory, and the instantaneous zero sequence harmonics in the time domain are extracted. By comparing the correlation of instantaneous zero sequence harmonics, we can get a correlation coefficient matrix and a comprehensive correlation coefficient. By comparing the difference between the maximum value and the minimum value of comprehensive correlation coefficients and the relationship between thresholds, we can select the fault line. The proposed method integrates transient state and steady state characteristics, considering the harmonic signal and the multi-correlation analysis. The simulation results show that the method is effective, and the physical meaning is clear.

Keywords: Instantaneous power, resonant grounded system, single-phase earth fault, Correlation, Fault line selection

1. Introduction

Because of the low impact of the single phase to ground faults in a resonant grounding system, the power supply reliability is high, so it is widely used in a distribution network in China. The operation mode of distributing networks is changeable, the structure is complex, and the zero sequence current is weak; therefore, it is difficult to detect the resonant grounding fault[1].

Domestic and foreign scholars have done a lot of work and have put forward a variety of resonance grounding fault line selection methods, such as zero sequence admittance method [2], wavelet analysis method [3], traveling wave detection method [4], signal injection method [5], etc. Because of only utilizing the partial failure signal, the fault feature of the signal is not obvious, and the reliability is not high. A method of line selections based on transient energy is proposed in the reference[6], which is based on the magnitude and direction of transient energy to determine the fault line, The method can accurately select the line with large transition resistance at the end of the line, when the transition resistance is very small; due to the proportion of an active component in the transient current is small, transient signals are not fully used, and the detection sensitivity is low. The correlation analysis is applied to the fault line selection in the reference [7], and according to the analysis of an integrated correlation degree of the transient current waveforms relative to feed circuits, the suite of detecting the faulty line is obtained and arranged by the size of the probabilities for grounding faults to occur. The adaptability of this method is good, but the computation is large. The method of line selection by using an instantaneous power theory is analyzed in the reference [8]. Through comparing instantaneous power in faulty lines and health lines before and after the fault, it is pointed out that the variation of instantaneous power in health lines before and after the fault is not evident; the variation of instantaneous power in faulty lines before and after the fault is evident. Under the condition of non sinusoidal, the instantaneous power theory considers the energy change of each harmonic, so it can accurately and actually describe the energy characteristics of the system. However, the line selection algorithm comparing the feature of the specific sub harmonic components does not maximum utilize the fault information.

*Corresponding Author: Shang Liqun
(E-mail: shanglq@xust.edu.cn)
College of Electrical and Control Engineering, Xi'an University
of Science and Technology, Yan-ta Rd., Xi'an, 710054,
China

Considering the above factors, in this paper, a new method of line selections based on instantaneous power theory and correlation algorithms is proposed, and it can fully utilize the steady state and transient state characteristics of fault signals. Based on the theory of instantaneous power and the comparison and the analysis of fault lines and the normal ones, the result shows that in the time domain, the fault lines have the different amplitudes with non fault line instantaneous zero sequence, and, moreover, they have the opposite directions. Therefore, we detect the fault line by a correlation analysis.

2. Basic Theory of Instantaneous Power

An instantaneous power theory discussed in the time domain has no limit to the waveforms of voltages and currents, which is suitable not only for the steady state, but also for the transient state. Being different from the traditional power concept, three-phase system as three single-phase systems are dealt with, and the three-phase system is dealt with as a unit to consider in the instantaneous power theory[9][10].

The instantaneous power theory adopts the Clarke transformation to transform the three-phase voltage and the three-phase current from the abc coordinate system to the $\alpha\beta 0$ stationary coordinate system where coordinates are perpendicular to each other, and the new definition of each component is then carried out in the $\alpha\beta 0$ coordinate system[11]. The change process is as follows:

Setting up the three-phase voltages and the three-phase currents of the circuit respectively are u_a, u_b, u_c and i_a, i_b, i_c . Using Clarke transformation to the three-phase voltage and the three-phase current respectively, we obtain the instantaneous voltage and current in the new $\alpha\beta 0$ coordinate system as follows:

$$\begin{bmatrix} u_0 \\ u_\alpha \\ u_\beta \end{bmatrix} = \sqrt{\frac{2}{3}} \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ 1 & -\frac{1}{2} & -\frac{1}{2} \\ 0 & \frac{\sqrt{3}}{2} & -\frac{\sqrt{3}}{2} \end{bmatrix} \begin{bmatrix} u_a \\ u_b \\ u_c \end{bmatrix}$$

$$\begin{bmatrix} i_0 \\ i_\alpha \\ i_\beta \end{bmatrix} = \sqrt{\frac{2}{3}} \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ 1 & -\frac{1}{2} & -\frac{1}{2} \\ 0 & \frac{\sqrt{3}}{2} & -\frac{\sqrt{3}}{2} \end{bmatrix} \begin{bmatrix} i_a \\ i_b \\ i_c \end{bmatrix} \quad (1)$$

We define the instantaneous zero sequence power, and the instantaneous power is based on this:

$$p_0 = u_0 i_0 \quad (2)$$

$$p = u_\alpha i_\alpha + u_\beta i_\beta \quad (3)$$

$$q = u_\beta i_\alpha - u_\alpha i_\beta \quad (4)$$

In the above equation, p_0 represents the instantaneous zero sequence power, p and q represent the instantaneous real power and the instantaneous imaginary power, respectively; the unit is "Fu Anxu(Vai)" [12-14].

The detection method based on an instantaneous reactive power theory is widely used because of its good performance and wide application ranges. It can be a complete analysis and real-time calculation of the different powers and the corresponding currents in the three-phase circuit [10][15]. The core idea is to distinguish the current component from the circuit in real time so as to eliminate it. Based on this idea, a method of $i_p - i_q$ for detecting harmonics and reactive currents is studied.

$i_p - i_q$ detection method does not directly use the three-phase voltage to participate in the calculation, but using sinusoidal signal $\sin \omega t$ and cosine signal $\cos \omega t$. A simulation diagram of $i_p - i_q$ harmonic detection method is shown in Figure 1. From Equation (5) we can calculate i_p, i_q according to the definition. After the low-pass filter to filter the AC components in the i_p, i_q to get the DC components, $\bar{i}_p, \bar{i}_q$, then $\bar{i}_p, \bar{i}_q$ are plugged into the Equation (6). And the fundamental components, i_{af}, i_{bf}, i_{cf} of the detected current are calculated, Three-phase currents, i_a, i_b, i_c minus i_{af}, i_{bf}, i_{cf} are harmonic currents.

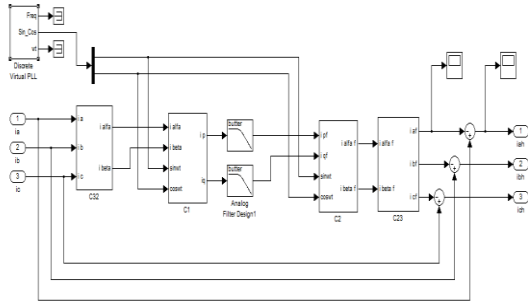


Figure 1: The simulation diagram of $i_p - i_q$ harmonic detection method

$$\begin{bmatrix} i_p \\ i_q \end{bmatrix} = CC_{32} \begin{bmatrix} i_a \\ i_b \\ i_c \end{bmatrix} = C \begin{bmatrix} i_\alpha \\ i_\beta \end{bmatrix} \quad (5)$$

Where, $C = \begin{bmatrix} \sin \omega t & -\cos \omega t \\ -\cos \omega t & -\sin \omega t \end{bmatrix}$

$$\begin{bmatrix} i_{df} \\ i_{bf} \\ i_{cf} \end{bmatrix} = C_{23} C \begin{bmatrix} i_p \\ i_q \end{bmatrix} = C_{23} \begin{bmatrix} \sin \omega t & -\cos \omega t \\ -\cos \omega t & -\sin \omega t \end{bmatrix} \begin{bmatrix} i_p \\ i_q \end{bmatrix} \quad (6)$$

For $i_p - i_q$ method in an instantaneous voltage is not directly involved in the computation, and only provides the sin and cos values to complete the calculation; voltage distortion components does not appear in the operation of the process. When the grid voltage is unbalanced or distorted, $i_p - i_q$ method is used to detect the phase voltage and frequency, and a three-phase current signal is used to real-time computation to detect harmonic currents; therefore, even the three-phase electrical signal is distorted or unbalanced, it can also accurately detect three-phase harmonic currents[16].

3. Principle of Line Selection in a Resonant Grounding System

3.1 Harmonic Current Detection Method

When the single phase to an earth fault occurs in the resonant grounding system, the zero sequence signal is present in the system. Besides the fundamental signal, the harmonic component and transient component are also included. The harmonic signal is mainly 5 or 7 harmonics. Since the 5 or 7 harmonic content of the system is limited, the fault criterion is not obvious, affecting the result of the fault line selection. By extracting the harmonic wave, the harmonic information of a single phase to ground fault can be fully used to avoid the shortcomings of the single harmonic content, and it is expected to improve the reliability of fault line selections.

The distribution of the system current is shown in Figure 2 where the single phase to ground fault occurs in the resonant grounding system.

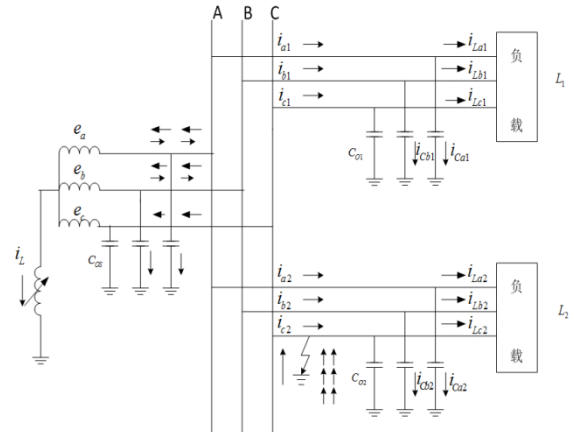


Figure2: The schematic diagram of a single phase earth fault in the resonant grounding system

As far as the sound circuit is concerned, the zero sequence current flowing through the circuit is determined by its own grounding capacitance. Considering the fault characteristic of the fundamental component, the fault line is a single phase earth fault; whether it is a fault line or a non-fault line, the three-phase current contains the sum of fundamental positive sequences, negative sequence components and the fundamental zero sequences, positive sequence harmonic, harmonic and negative sequences and harmonic zero sequence components, expressed as follows :

$$\begin{cases} i_a = i_{Ca} + i_{La} = i_{af_+} + i_{af_-} + i_{af_0} + i_{an_+} + i_{an_-} + i_{an_0} \\ i_b = i_{Cb} + i_{Lb} = i_{bf_+} + i_{bf_-} + i_{bf_0} + i_{bn_+} + i_{bn_-} + i_{bn_0} \\ i_c = i_{Cc} + i_{Lc} = i_{cf_+} + i_{cf_-} + i_{cf_0} + i_{cn_+} + i_{cn_-} + i_{cn_0} \end{cases} \quad (7)$$

In the above equation, i_a, i_b, i_c are three-phase currents that flow through the head of the line. i_{Ca}, i_{Cb}, i_{Cc} are three-phase to ground capacitive currents, which are produced by the phase voltage. i_{La}, i_{Lb}, i_{Lc} are load currents at the end of the line, which are produced by the line voltage. i_{af_+} is the fundamental positive sequence current; i_{af_-} is the fundamental negative sequence current; i_{af_0} is the fundamental zero sequence current; i_{an_+} is the positive sequence harmonic current, i_{an_-} is the harmonic negative sequence current, and i_{an_0} is the harmonic zero sequence current.

By using the instantaneous power theory, the fault signal is processed. The fundamental positive sequences, negative sequence components and the fundamental zero sequence components are detected in turn. The fundamental component and Original three-phase current subtraction, and the harmonic component of three-phase current can be obtained, namely:

$$\begin{cases} i_{ah} = i_a - (i_{af_+} + i_{af_-} + i_{af_0}) \\ i_{bh} = i_b - (i_{bf_+} + i_{bf_-} + i_{bf_0}) \\ i_{ch} = i_c - (i_{cf_+} + i_{cf_-} + i_{cf_0}) \end{cases} \quad (8)$$

From Equation (1) we can obtain the relationship between instantaneous zero sequence components and three phase currents, and extracted zero sequence components from three phase harmonics that can be obtained from zero sequence harmonic currents of each branch.

$$i_{h_0} = \frac{1}{\sqrt{3}}(i_{ah} + i_{bh} + i_{ch}) \quad (9)$$

However, if we want to extract in three-phase current harmonic component, at first, we need to filter out the current fundamental component, which due to the asymmetry, it starts from the angle of the order, including positive sequence fundamental, negative sequences and zero sequence components. Based on the instantaneous power theory, $i_p - i_q$ method, in fact, is used to obtain the instantaneous line active power and instantaneous no power to filter the oscillating part, and retain with the positive sequence fundamental current corresponding to the average component, and inverse transform detect of fundamental positive sequence component of the three-phase current. According to this idea, three-phase current fundamental positive sequences, negative sequences and zero sequence currents are detected.

3.2 Line Selection Algorithm

The correlation function is a kind of very important digital characteristics used to describe a series of random signal statistical characteristics according to the concept and the basic definition, it is also applied to deterministic and uncertain signals [17]. The correlation analysis method is a kind of important signal processing tool, and it can be used to study two signal similarity, or to describe a signal in a period of time before and after the self similarity, so as to realize the detection and extraction of the signal.

The definition of the cross correlation function of the two signals $x(t)$ and $y(t)$ is as follows:

$$R_{xy}(\tau) = \frac{1}{T} \int_0^T x(t)y(t+\tau)dt \quad (10)$$

In the formula, T is the average time, and τ is the time difference between the two signals.

By Equation (10), it can be seen that the cross correlation function of the essence is the period average of two signals, where a signal $x(t)$ can lag or lead another signal $y(t)$ τ s. τ zero represents the signal without time differences. This will clearly show the similarity between random or stochastic signals, or the time relationship between two identical or similar signals.

The correlation analysis can synthesize the phase relation and amplitude information of the two signals, but not the amplitude of the single frequency[18]. After three-phase current treatment, the instantaneous zero sequence harmonic current of each line after single-phase grounding fault is obtained. And taking this as the fault characteristic signal, and carrying on the correlation analysis, it can comprehensively compare the correlation between two signals.

4. Simulation Verification

MATLAB simulation model is shown in Figure 3. The system is for the hybrid cable, and the resonant grounding system model with 5 outgoing lines, and then sets up 5 outgoing lines' length being 10km, 9km, 8km, 5km and 3km, respectively. Joining harmonic sources in simulation and harmonic frequencies for 5 times and 7 times, the sizes are 2% and 7% of fundamental current, respectively. The failure time is set in 0.02s.

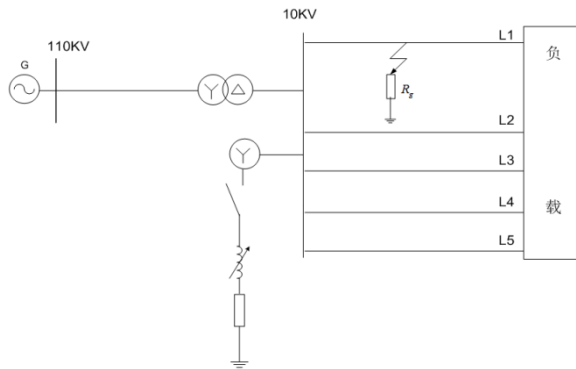
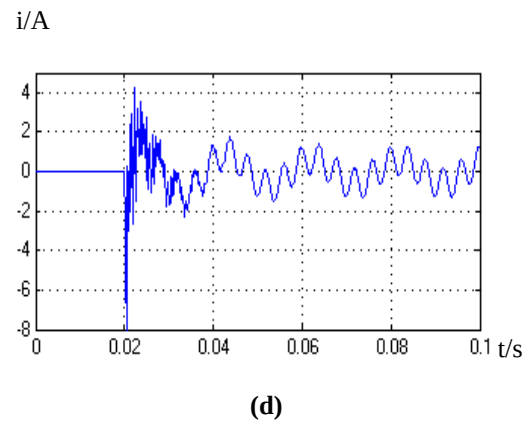
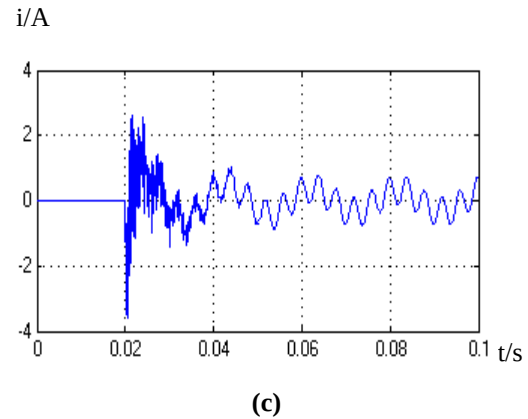
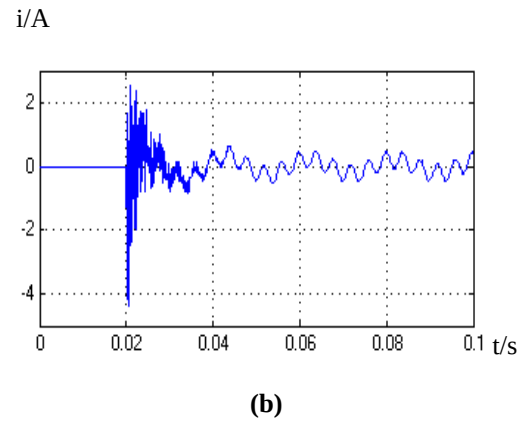
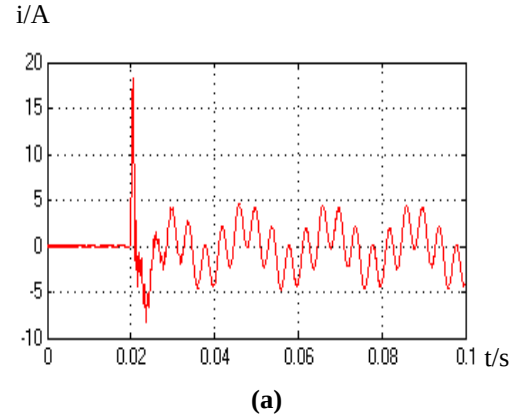


Figure 3: The simulation model diagram

First of all, the compensation degree is 8% over the compensation mode; and set the fault occurred in line 1 of the first section, the fault initial angle of 0° , and ground resistance of 300Ω . Each line of instantaneous zero sequence harmonic waves is shown in Figure 4 after fault.



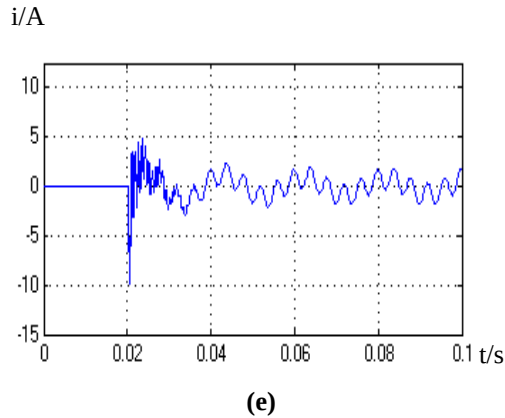


Figure 4: Instantaneous zero sequence harmonic waves of lines. (a) Fault line 1(b) Non fault line 2(c) Non fault line 3(d) Non fault line 4(e) Non fault line 5

From Figure 4 we can see, when it is 0.02s, the circuit has a single-phase grounding fault and there is a great impact on currents, but the transient process is relatively short, about in 1 / 4 cycle, and it tends to be steady state. After a fault line, a fault transient zero sequence harmonic current in the process of transient and steady state amplitude is bigger than the other four non-fault lines and has same phase contrast, and a different waveform. However, the amplitude and phase of other four non-fault lines are the same, and the waveforms are similar. It shows that the instantaneous zero sequence harmonic current wave forms of a single phase to ground fault line and non-fault line are significantly different.

According to the comprehensive calculation of correlation coefficients of each line, we can get the comprehensive correlation coefficient matrix E :

$$E = [-0.7280 \quad 0.3410 \quad 0.3945 \quad 0.4021 \quad 0.3778]$$

From the above correlation coefficient matrix E , we can see the minimum -0.7280 element of the first column has the big difference with the other four elements, but the other four elements are similar. By calculating, we can get $p = 1.0690 > p_{set}$ and choose line 1 for the fault line.

Different lines have different faults, and the state and different fault line selection of the results are shown in the table below. You can see that the comprehensive line selection method is based on instantaneous power, and single-phase earth faults

occurred in different lines and different fault cases, so we can check out fault lines correctly.

Table 1: Line selection result of different fault states and different lines

Fault line e	Fault parameter	p number	Result
l_1	$X = 10km$ $R = 500\Omega$ $\theta = 45^\circ$	1.1466	l_1
l_3	$X = 0km$ $R = 50\Omega$ $\theta = 90^\circ$	1.2655	l_3
l_4	$X = 2.5km$ $R = 500\Omega$ $\theta = 90^\circ$	1.3862	l_4

5. Conclusions

When the single phase to a ground fault occurs in the resonant grounding system, the fault signal is more complex, and the non sinusoidal characteristic is highlighted by the addition of harmonics and interference. The frequency characteristics of fault signals can be accurately described by the instantaneous zero sequence power. When a fault occurs, fault line and non-fault line zero sequence harmonics have the different components; by comparing the correlation to analyze the harmonic current phase and amplitude information, each line in the comparison of the instantaneous zero sequence harmonic current is similar. Therefore, we can detect the fault line accurately. This method does not need to calculate the phase, considering that both the steady-state fault characteristics and the transient fault information, and the multiple criteria are complementary. Theoretical analysis and simulation results show that the proposed method is effective.

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Wu Xing is currently a graduate student at Xi'an University of Science & Technology, P. R. China. His main areas of interest are power system control and protection.